Topic to be tested

- Lecture 2 Algorithm Analysis
  - Complexity, Big-O notation
- Lecture 9 AVL Trees
  - Height V.S. Size (the number of nodes), Insertion
  - Deletion will NOT be tested
- Lecture 10 Priority Queue (Heaps) & Heap Sort
  - Heap Property, Insertion
  - Deletion will NOT be tested

Topic to be tested

- Lecture 11 Hashing
  - Hash table, hashing, hash function
  - Collision handling
    - separate chaining, 3 open address methods
- Lecture 12-1 & 12-2 Sorting
  - Sorting algorithms
  - Analysis, esp. that of merge sort and quick sort (the analysis of running time of recursive routines)
- Lecture 13 & 14 Graph
  - Terminologies, Topological sort, BFS, DFS

Format

- Question-Answering Questions
- 2 hours
- Close-book/notes exam
- Use of calculator is allowed
- C programming will NOT be tested.
- But text description/pseudocode of algorithms may be tested.
- Please focus on the slides and exercises of the aforementioned lectures.